Synthetic EEG Signal Generation for Motor Imagery Tasks Using Conditional Variational Autoencoder (CVAE)

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CSC 550

May 2025

Github

Introduction

Electroencephalography (EEG) signals are a cornerstone of non-invasive Brain-Computer Interface (BCI) technologies, which enable users to control external systems, for example, prosthetics or communication aids, by imagining specifics motor tasks. However, acquiring EEG data is fraught with challenges: It's expensive, time-consuming, prone to noise, and subject to strict ethical and privacy constraints. To address these issues, we have attempted to develop a conditional variational autoencoder (CVAE) that can generate artificial EEG signals for four motor imagery task:

- left and right hand movement
- feet movement
- tongue movement

The goal is to enhance training dataset for BCI models while preserving user privacy. The figure below are examples of the EEG data of patients from the BCI Competition IV Dataset 2a.

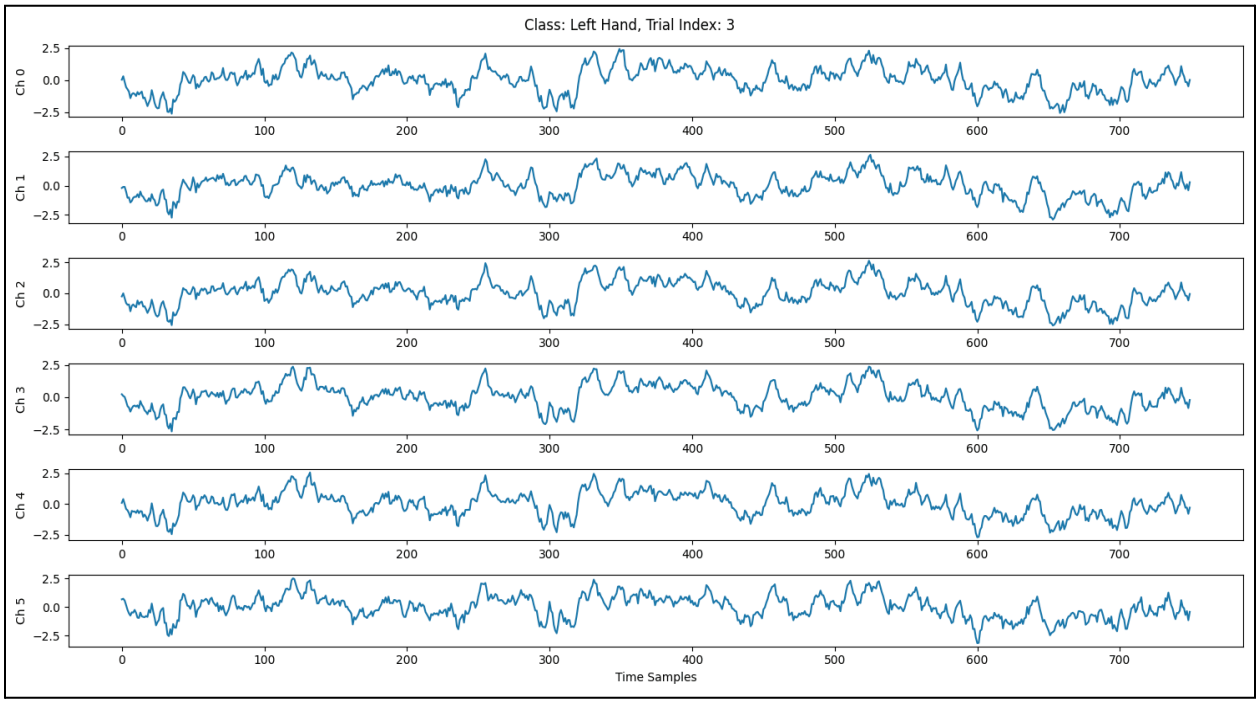


Figure 1. left hand, trial index 3, Figure 2,3, and 4 contain the other motor images in Appendix

Methods

Dataset & Preprocessing

We utilized the BCI Competition IV Dataset 2a, comprising EEG recordings from nine subjects performing motor imagery tasks, sampled at 250 Hz across 22 channels. Raw EEG was extracted using custom MATLAB compatible Python tools. Each trial was segmented into 3-second windows (750 samples) aligned with task onset. Data was normalized using z-score normalization on a per-channel, per-trial basis.

Model Architecture

The CVAE architecture includes two fully connected encoder layers (512 and 256 units) with ReLU activations and dropout, compressing inputs into a 64-dimensional latent space. Decoder layers mirror the encoder, reconstructing inputs conditioned on task labels. Novel contributions include:

- **L1 regularization** for signal sparsity
- A **smoothness penalty** to encourage physiologically plausible EEG dynamics
- Dropout layers to reduce overfitting
- Combines **Mean Squared Error (MSE)** for reconstruction quality, **Kullback-Leibler (KL) divergence** for latent space regularization, **L1 regularization** to promote signal sparsity, and a **smoothness penalty** to mimic physiological plausibility.

Training Strategy

We employed 5-fold cross-validation to evaluate generalization and prevent overfitting. To further improve robustness, we implemented early stopping with a patience of five epochs based on validation loss. Additionally, input sequencees were truncated to their central 50%, focusing the model on the most informative EEG segments. The CVAE was trained using the Adam optimizer with a learning rate of 0.001, batch size of 64, and a maximum of 50 epochs

Results

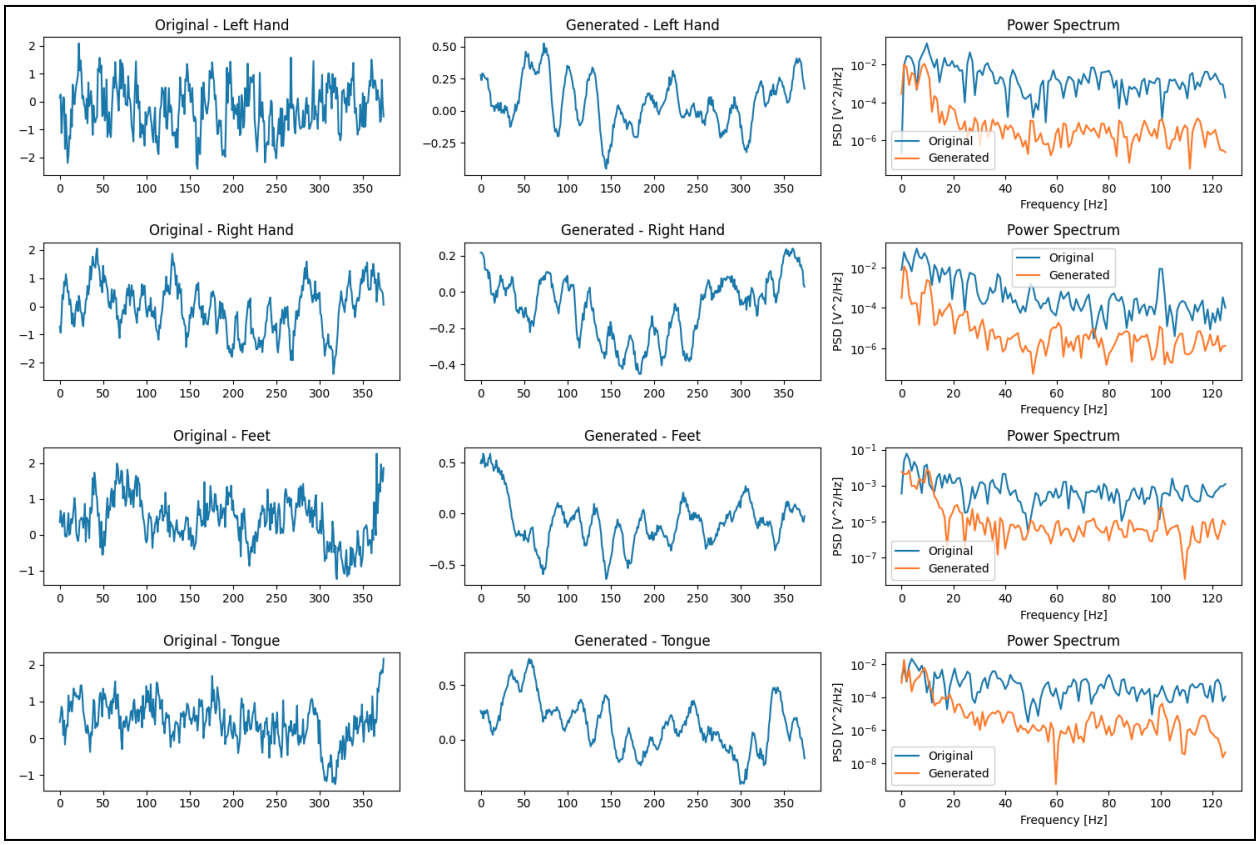


Figure 7:.Time and Frequency Comparison of Real vs. Generated EEG Signals Across Motor Imagery Classes

This shows a comparison between real and generated EEG signals across all four motor imagery classes. The generated signals follow a similar overall structure in the time domain, capturing specific patterns like those seeen in left vs. right hand imagery. In the frequency domain, the synthetic signals also retain the dominant low-frequency components (under 30 Hz), but show less power in the higher frequency range, something we noted. Despite this, the signals are consistent and distinct enough to suggest that the model is learning useful class-conditioned features.

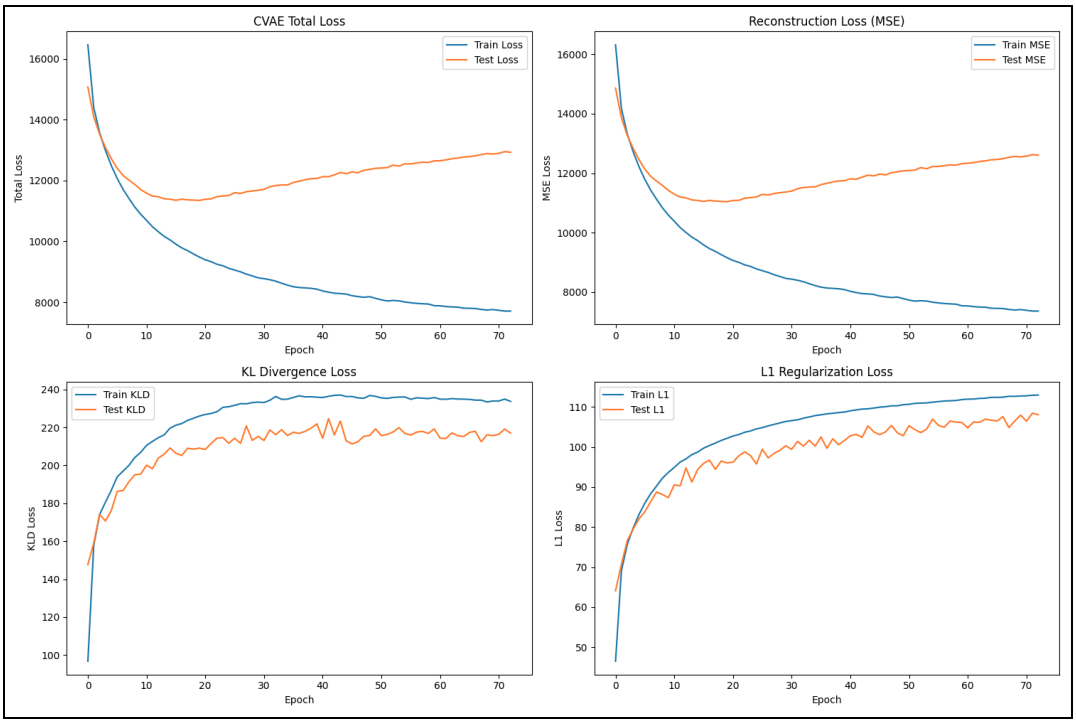


Figure 5. Training and Test Loss Curves – Initial CVAE Implementation

In the initial training configuration (Figure 1), the CVAE exhibited clear signs of overfitting. While the training loss decreased consistently across all components—including total loss, reconstruction error (MSE), KL divergence, and L1 regularization—the test loss diverged significantly after approximately epoch 20. The reconstruction loss plateaued on the test sets, while the KLD adn L1 terms continued to increase, suggesting that the model over-optimized on training data distributions at the expense of generalization.

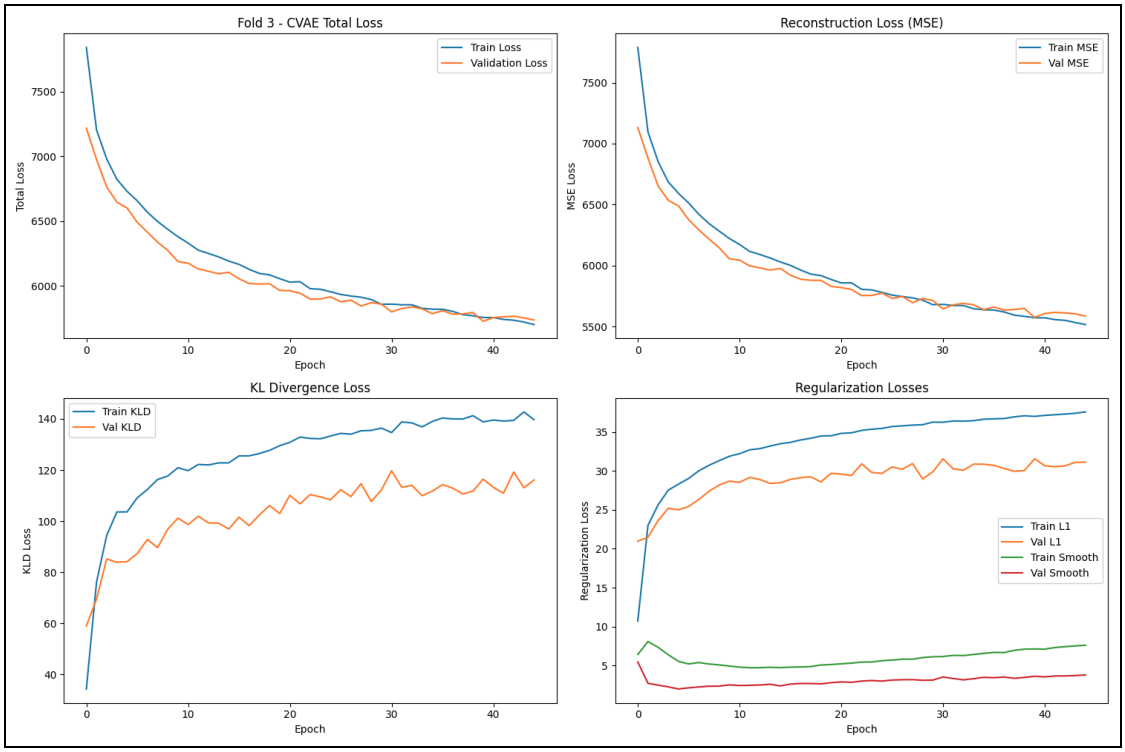


Figure 6. Training and Validation Loss Curves – Fold 3 (Cross-Validated CVAE)

After receiving feedback from the professor and TAs, we introduced 5-fold cross-validation, early stopping, and a smoothness penalty, with Fold 3 emerging as the best-performing model (Figure 2). The learning curves for this fold demonstreate a much tighter alignment between training and validations losses, with no visible overfitting.

- The total loss and reconstruction error converge smoothly over 50 epochs.
- The KL divergence remains relatively stable across training and validation sets, indicating consistent latent structure.
- The newly introduced smoothness regularization maintained a steady profile, and the L1 penalty was well-balanced across both sets, further supporting the model's generalization.

Visualizations revealed that the generated EEG signals:

- Matched the structural morphology of real EEG signals.

- Retained specific characteristics
- Captured dominant frequency components (<30 Hz), although with less high-frequency content than real data.

Discussion

In this project, we explored how Conditional Variational Autoencoders (CVAEs) can be used to generate synthetic EEG signals based on motor imagery tasks. The generated signals ended up looking and behaving a lot like real EEG data, especially in terms of their overall shape over time and their frequency patterns below 30 Hz. This is important because collecting high-quality EEGs data can be expensive and time-consumings so having a model that can generate realistic examples could really help with data augmentation in BCI research.

After noticing overfitting in our first model, we improved the training setup by adding cross-validation, focusing on the middle portion of each EEG trial, and including a smoothness penalty. These changes helped the model generalize better, and we saw much closer alignment between training and validation losses. Overall, this shows that a mix of good preprocesssing and regularization techniques can make a big difference when training generative models on EEG signals.

Limitations & Future Work

While our CVAE model performed well in capturing the dominant low-frequency components of EEG signals, it struggled to replicate high-frequency signal fidelity, which can be critical for certain cognitive and motor-related EEG tasks. This limitation may impact the usefulness of the generated data in applications that rely on fine-grained detail.

To improve performance, future work will explore the incorporation of frequency-domain constraints directly into the loss function, encouraging the model to match the properties of real EEG more closely. We also plan to experiment with EEG-specific data augmentation techniques, such as amplitude scaling and channel swapping, to help improve generalization and robustness.

Conclusion

In this project, we built a Conditional Variational Autoencoder that can generate synthetic EEG data based on different motor imagery tasks. Our approach helps tackle some common challenges in BCI research, like limited access to large datasets, privacy concerns, and poor generalization. While the results are promising, there's still room to improve, especially in making the generated signals even more realistic. In the future, we hope to expand this work to make it more useful for real-world applications in neurotechnology and assistive healthcare.

References

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Khalid, Z., & Joseph, N. (2025). *CVAE for Synthetic EEGs* [our implementation]. GitHub. https://github.com/xZaisk/CVAE-for-Synthetic-EEGs

Yue, Y., Deng, J. D., De Ridder, D., Manning, P., & Adhia, D. (2023, February 1). *Variational autoencoders learn better feature representations for EEG-based obesity classification*. arXiv.org. https://doi.org/10.48550/arXiv.2302.00789

Appendix

Figures 1-4

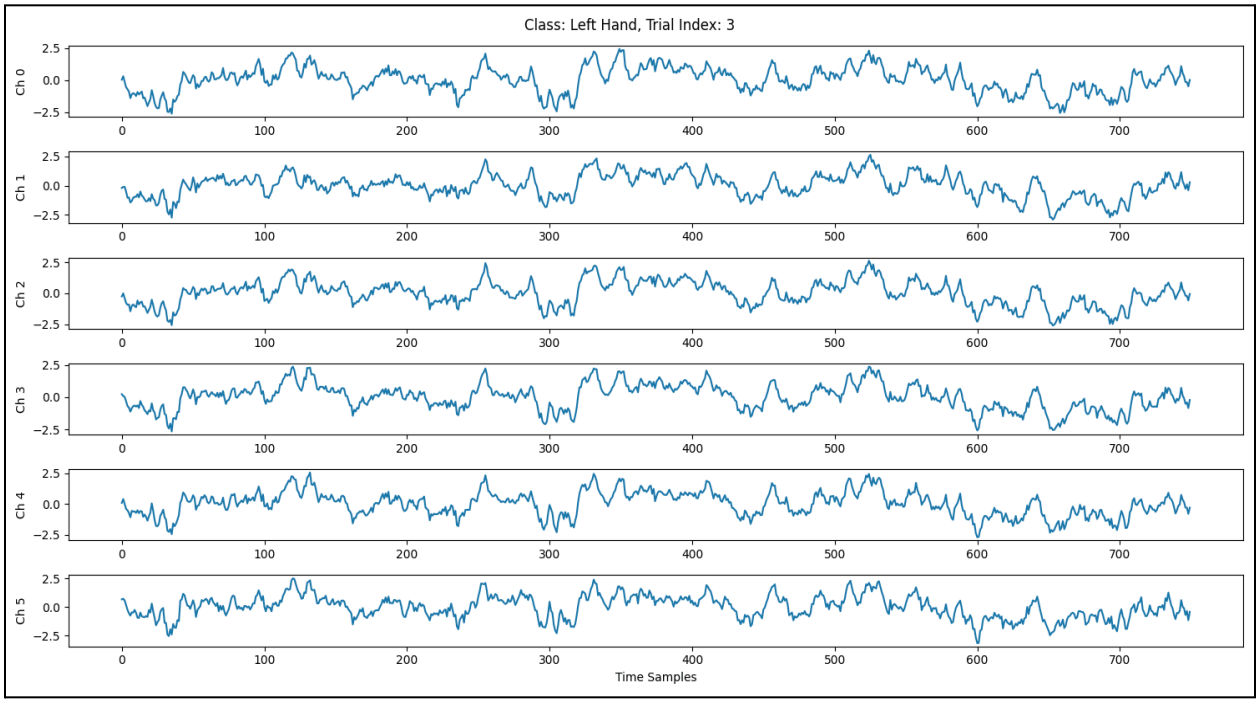


Figure 1. left hand, trial index 3

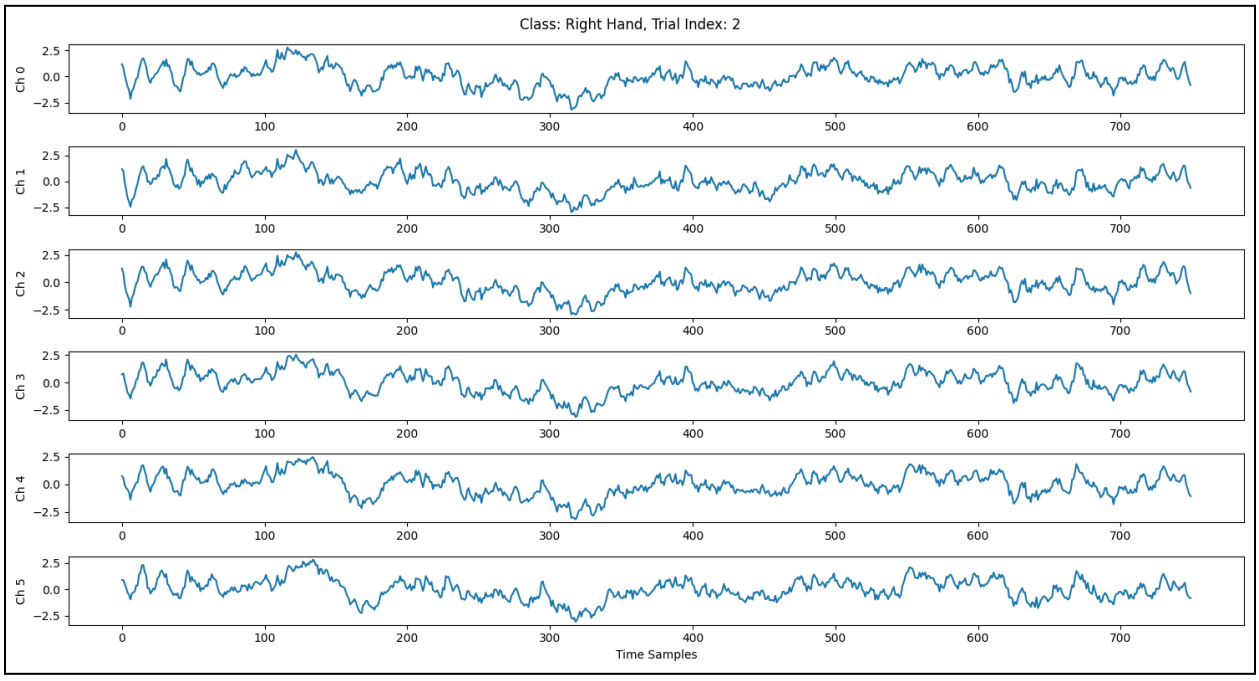


Figure 2. right hand, trial index 2

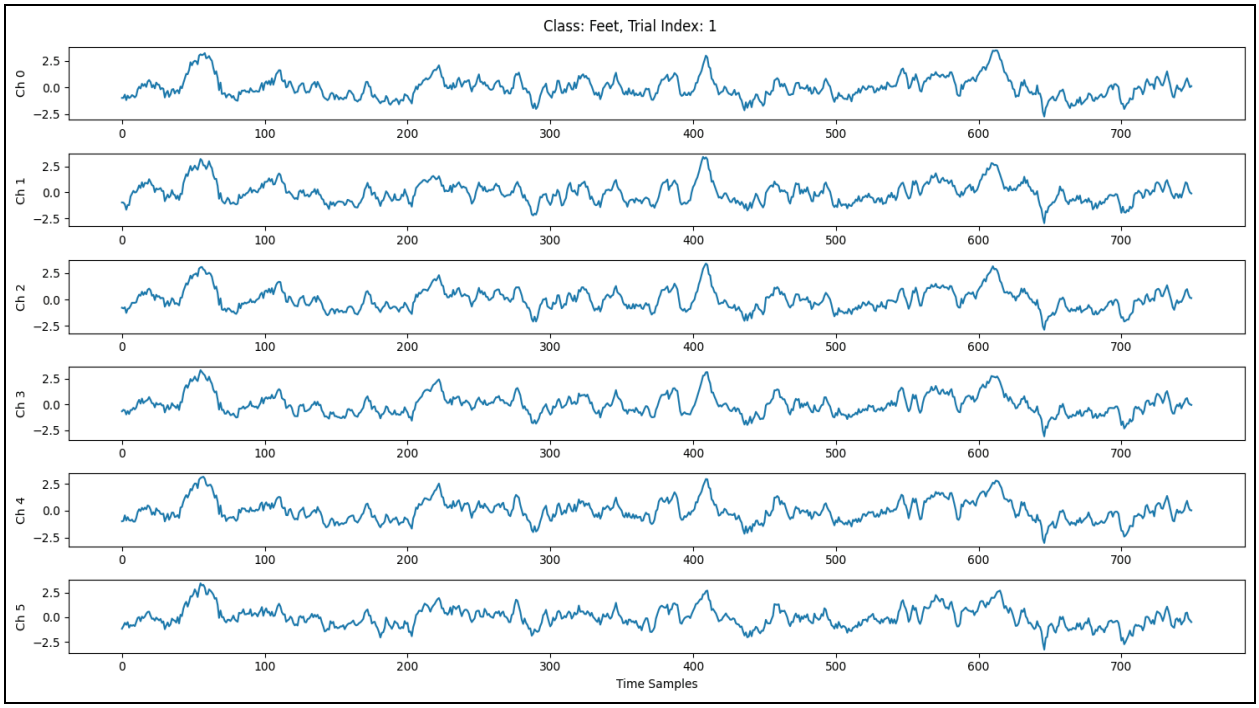


Figure 3. feet, trial index 1

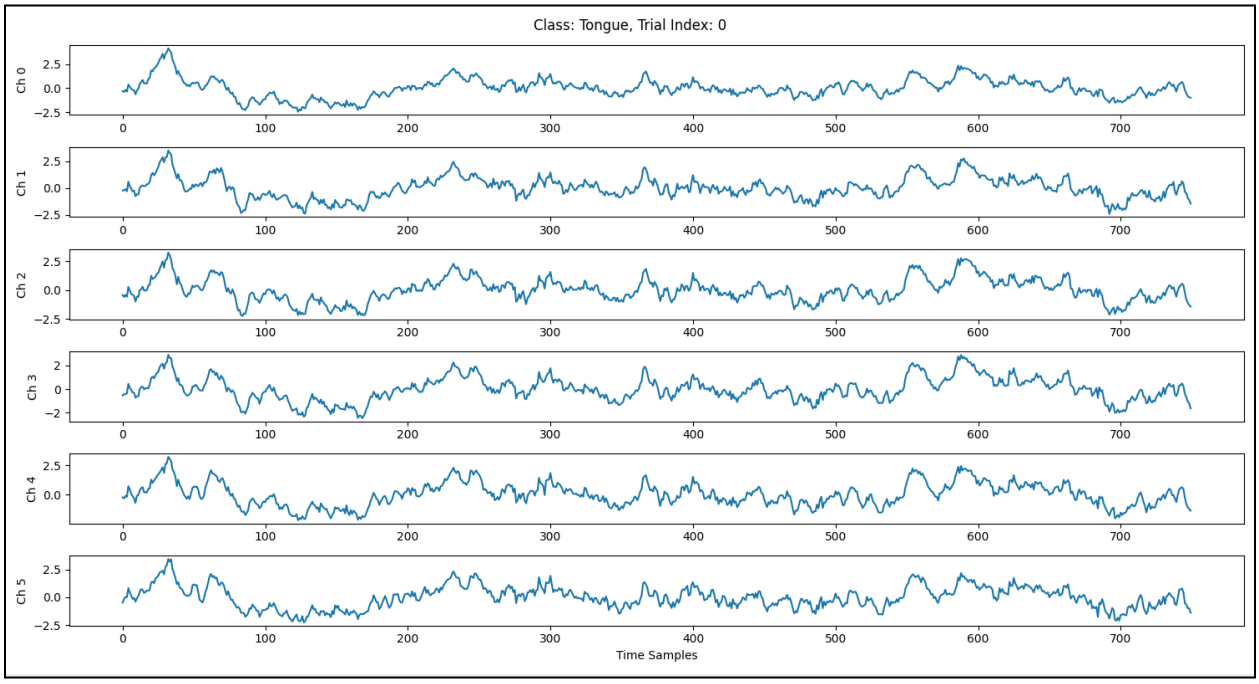


Figure 4. tongue, trial index 0

Result figures:

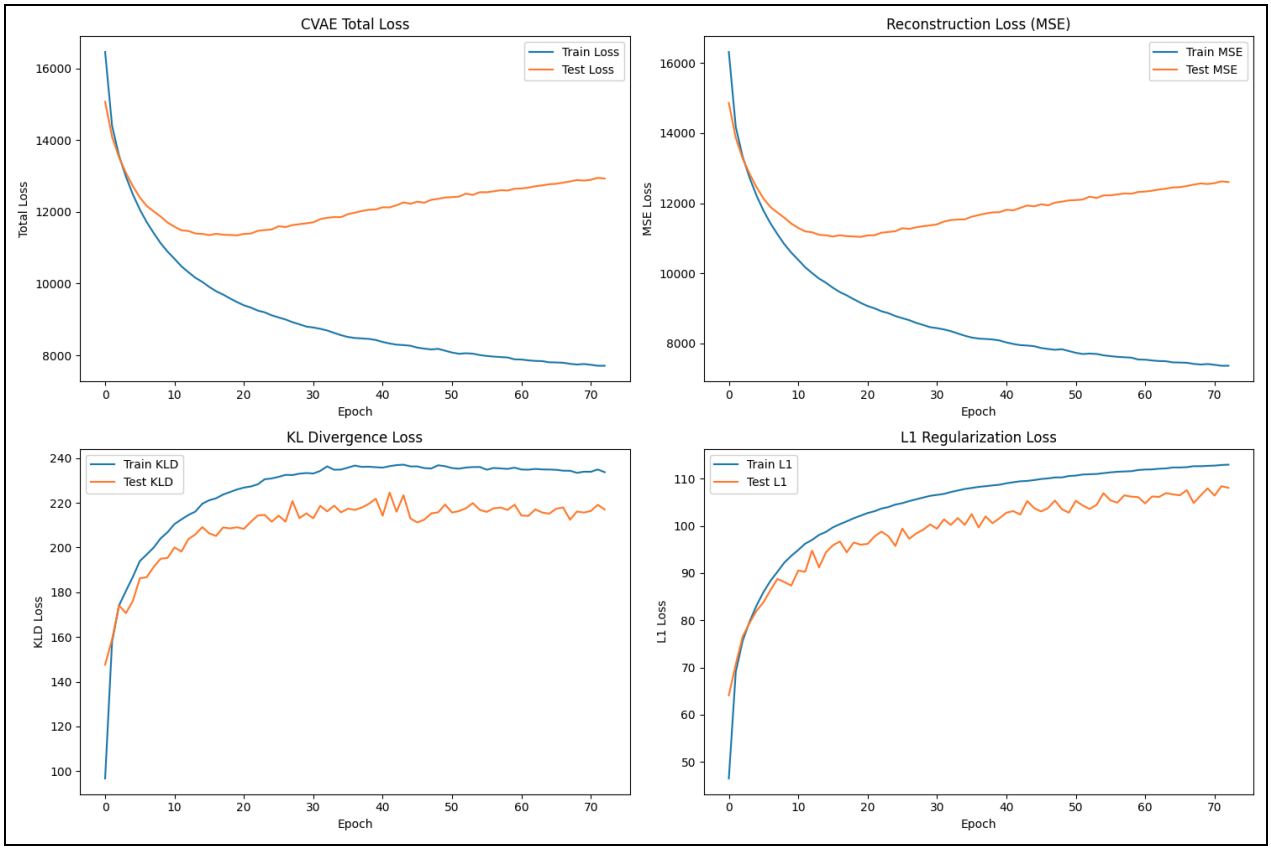


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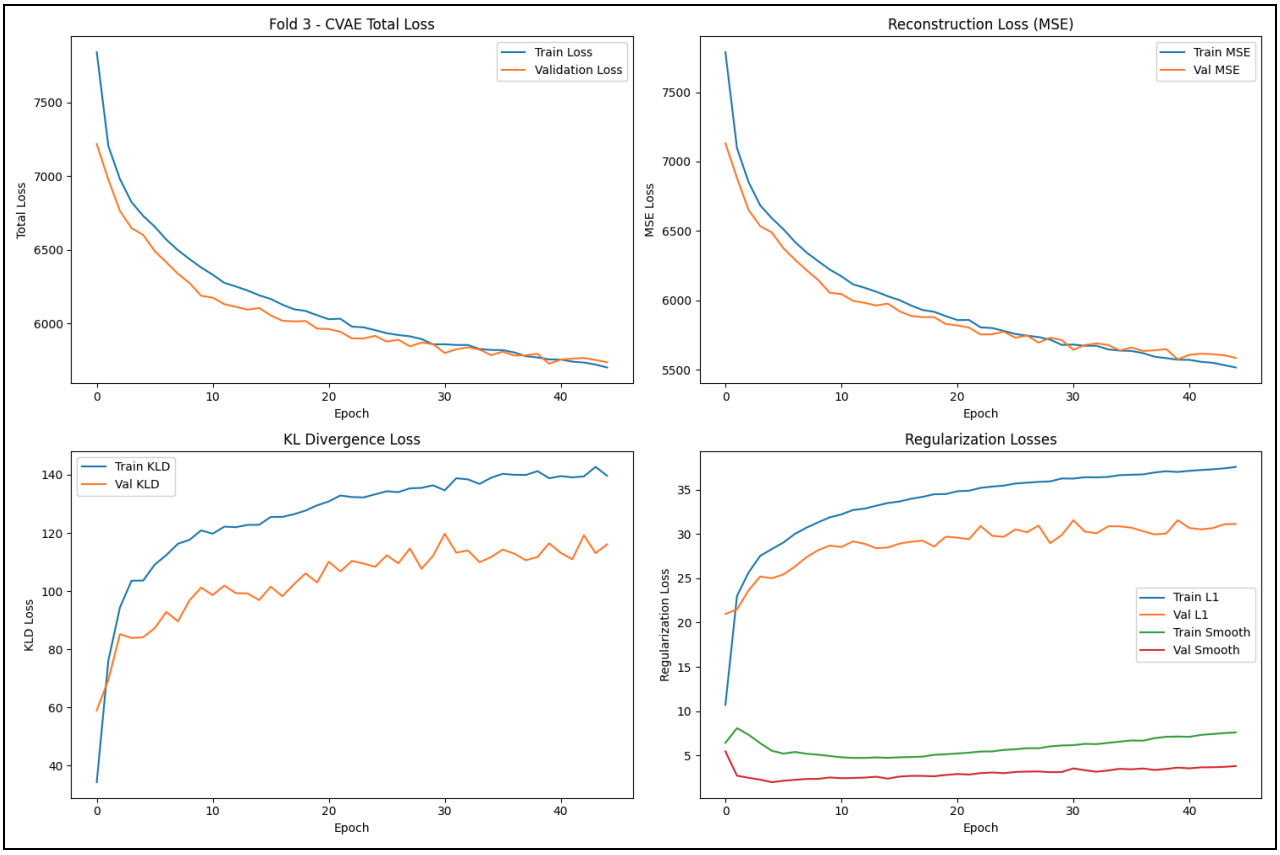


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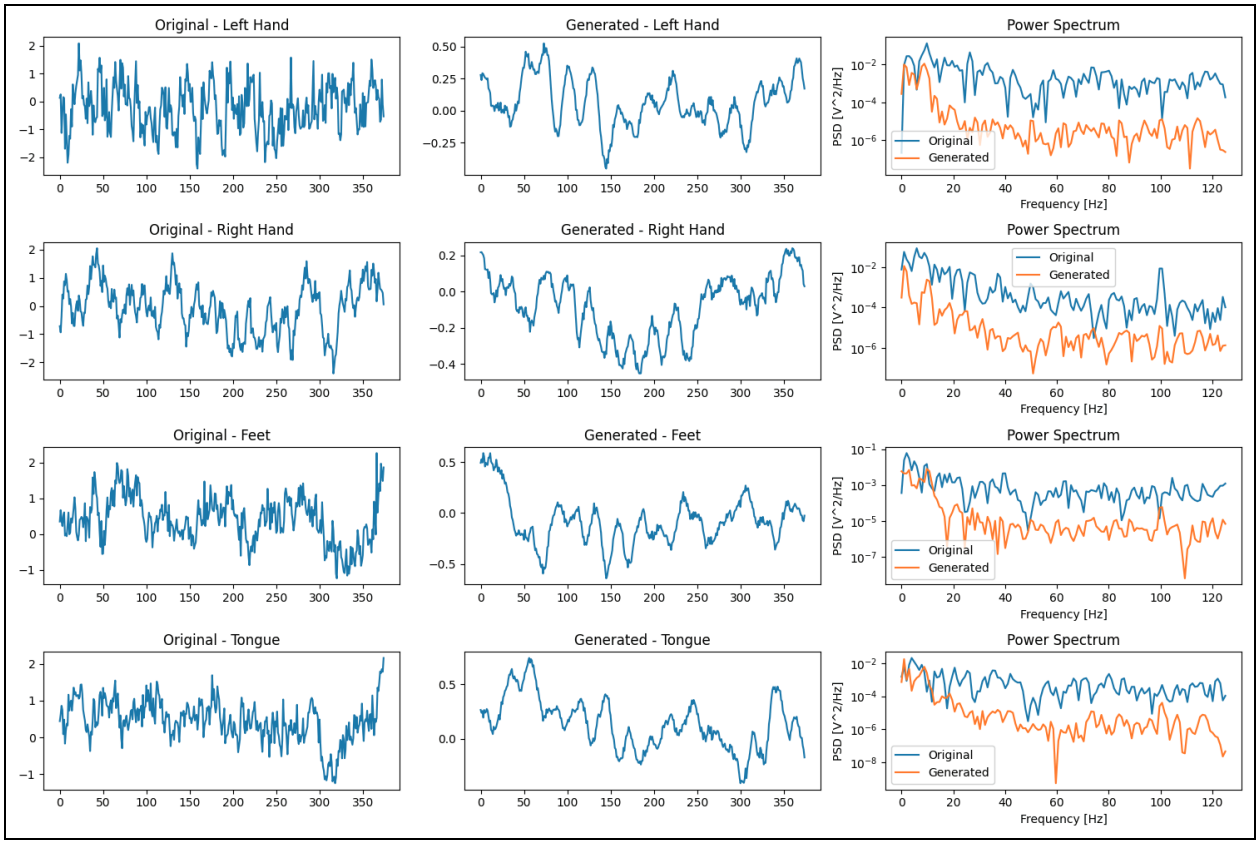


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